

Lecture 5: Solving Linear Equations and Null Space

MATH203 Applied Linear Algebra

Motivation: Why We Need to Solve Equations

Recall: The Two Languages (Lecture 4)

Every subspace has **two complementary descriptions**:

Language	How it works	Example
Descriptive	List equations (constraints)	$\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$
Constructive	Provide recipe (generate members)	$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots\}$

Today's question: What is each language **good at**? What is each language **bad at**?

Each Language Has Strengths and Weaknesses

	Descriptive $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$	Constructive $\text{span}\{\mathbf{v}_1, \dots\}$
Generation (make a member)	Hard Can't tell which $\mathbf{x}$ works	Easy Just pick coefficients
Verification (check membership)	Easy Just compute $A\mathbf{x}$	Hard Solve equation

Goal of this lecture: Learn to **solve equations** so we can:

- Convert descriptive $\rightarrow$ constructive (generation power)
- Convert constructive $\rightarrow$ descriptive (verification power)

Concrete Example: Finding Null Space

Example Setup: Shinchin's Coffee Shop Menu

Shinchin has three drinks on his menu:

	 milk	 coffee	 tea
 tea leaves	0	0	2
 lemon	0	0	1
 beans	0	2	4
 milk	1	0	1

Matrix form:

$$\underbrace{\begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 4 \\ 1 & 0 & 1 \end{pmatrix}}_A$$

The Question We Want to Answer

Question in coffee shop language: Which drinks can be made from other drinks?

For example:

- Can  be made from some combination of  and ?
- If so, what is the recipe?

Mathematical translation: We're looking for **relationships between the columns** of A .

If there's a relationship like “2 coffees = 1 tea”, this means:

$$2 \times \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix} = 1 \times \begin{pmatrix} 2 \\ 1 \\ 4 \\ 1 \end{pmatrix}$$

Setup: The Equation to Solve

We're looking for all vectors $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ such that:

$$\underbrace{\begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 4 \\ 1 & 0 & 1 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Strategy: Use **cross-filling** to decompose the augmented matrix $\underbrace{(A \mid \mathbf{0})}_{4 \times 4}$

into rank-one pieces, then solve backwards.

Step 1: Form Augmented Matrix

$$(A \mid \mathbf{0})$$

$$\left(\begin{array}{cccc} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 \\ 1 & 0 & 1 & 0 \end{array} \right)$$

Goal: Cross-fill to find rank-one pieces $(R_i \mid \mathbf{b}_i)$.

First step: Choose a pivot to start elimination.

Step 2: Choose First Pivot at (4,1)

$$\begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 \\ \leftarrow \text{column} & \boxed{1} & 0 & 1 & \rightarrow \text{center} \end{pmatrix} \quad \text{Pivot: } (4,1) = 1 \text{ (Red)}$$

Cross-filling formula:

$$\underbrace{(R_1 \mid \mathbf{b}_1)}_{\text{rank-one}} = \underbrace{\frac{\text{column 1}}{1}}_{\text{normalized column}} \cdot (\text{row 4})$$

Step 3: Extract First Rank-One Piece

$$\begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$(R_1 \mid \mathbf{b}_1)$ Remainder

Observation: Column 1 is now eliminated from the remainder!

Step 4: Choose Second Pivot at (3,2)

$$\left(\begin{array}{cccc} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ \text{col} \leftarrow 0 & \boxed{2} & 4 & 0 \\ 0 & \downarrow & 0 & 0 \\ \text{center} & & & \end{array} \right) \quad \text{Pivot: } (3,2) = 2 \text{ (Blue)}$$

Cross-filling formula:

$$\underbrace{(R_2 \mid \mathbf{b}_2)}_{\text{rank-one}} = \underbrace{\frac{\text{column 2 of remainder}}{2}}_{\text{normalized column}} \cdot (\text{row 3 of remainder})$$

Note: Column 2 of remainder = $\begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \end{pmatrix}$ (column 1 already zero)

Step 5: Extract Second Rank-One Piece

$$\begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{(R_2 \mid \mathbf{b}_2)} + \underbrace{\begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{\text{Remainder}}$$

Observation: Columns 1 and 2 are now eliminated! Remainder only has column 3 active.

Step 6: Choose Third Pivot at (1,3)

center
↑

$$\left(\begin{array}{cccc} \text{col} & \leftarrow 0 & \boxed{2} & \rightarrow 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Pivot: (1, 3) = 2 (Green)

Cross-filling formula:

$$\underbrace{(R_3 \mid \mathbf{b}_3)}_{\text{rank-one}} = \underbrace{\frac{\text{column 3 of remainder}}{2}}_{\text{normalized column}} \cdot (\text{row 1 of remainder})$$

Step 7: Complete Decomposition

$$\left(\begin{array}{cccc} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 \\ 1 & 0 & 1 & 0 \end{array} \right) = \left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{array} \right) + \left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) + \left(\begin{array}{ccc|c} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$(A \mid \mathbf{0}) = \underbrace{(R_1 \mid \mathbf{b}_1)}_{\text{Red}} + \underbrace{(R_2 \mid \mathbf{b}_2)}_{\text{Blue}} + \underbrace{(R_3 \mid \mathbf{b}_3)}_{\text{Green}}$$

Next: Read off the simplified equations from each piece!

Step 8: Read Off Simplified Equations

Each rank-one piece gives one simplified equation:

From $(R_1 \mid \mathbf{b}_1)$ (row 4):

$$1 \cdot x_1 + 0 \cdot x_2 + 1 \cdot x_3 = 0 \quad \Rightarrow \quad \boxed{x_1 + x_3 = 0}$$

From $(R_2 \mid \mathbf{b}_2)$ (row 3):

$$0 \cdot x_1 + 2 \cdot x_2 + 4 \cdot x_3 = 0 \quad \Rightarrow \quad \boxed{2x_2 + 4x_3 = 0}$$

From $(R_3 \mid \mathbf{b}_3)$ (rows 1,2):

$$\begin{cases} 0 \cdot x_1 + 0 \cdot x_2 + 2 \cdot x_3 = 0 \\ 0 \cdot x_1 + 0 \cdot x_2 + 1 \cdot x_3 = 0 \end{cases} \quad \Rightarrow \quad \boxed{x_3 = 0}$$

Key observation: Each equation involves **fewer variables** than the previous!

Step 9: Solve Backwards

We have three equations with decreasing complexity:

1. From $(R_3 \mid \mathbf{b}_3)$: $x_3 = 0$
2. From $(R_2 \mid \mathbf{b}_2)$: $2x_2 + 4x_3 = 0$
3. From $(R_1 \mid \mathbf{b}_1)$: $x_1 + x_3 = 0$

Solve backwards (start from the simplest):

$$x_3 = 0 \quad (\text{from equation 1})$$

$$2x_2 + 4(0) = 0 \quad \Rightarrow \quad x_2 = 0 \quad (\text{from equation 2})$$

$$x_1 + 0 = 0 \quad \Rightarrow \quad x_1 = 0 \quad (\text{from equation 3})$$

Conclusion: The only solution is $\mathbf{x} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.

Step 10: Interpret the Answer

Mathematical conclusion: The null space is:

$$\text{Null}(A) = \{\mathbf{0}\}$$

Coffee shop interpretation: There are **NO non-trivial relationships** between the drinks!

-  cannot be made from  and 
-  cannot be made from  and 
-  cannot be made from  and 

All three drinks are **linearly independent** $\Rightarrow$ no drink is redundant!

What We Just Did

Summary of the cross-filling method:

1. Form augmented matrix $\underbrace{(A \mid \mathbf{0})}_{m \times (n+1)}$
2. Decompose into rank-one pieces:
 $(A \mid \mathbf{0}) = (R_1 \mid \mathbf{b}_1) + (R_2 \mid \mathbf{b}_2) + \dots$
3. Each piece gives a simplified equation
4. Variables decrease with each step
5. Solve backwards from simplest to most complex

Now we formalize: What did we actually find? What is "null space"?

Null Space: Definition (Extracted from Example)

Definition (Null Space (Kernel))

For a matrix $\underbrace{A}_{m \times n}$, the **null space** is:

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

Meaning: $\text{Null}(A)$ records all **relationships between the columns** of A .

Also called the **kernel**, denoted $\ker(A)$.

In coffee shop language: Drink combinations that reveal "which drinks can be made from others."

In our example: $\text{Null}(A) = \{\mathbf{0}\}$ means no drink can be made from others.

Why Augmented Matrix $(A \mid \mathbf{0})$?

Question: Why did we form $(A \mid \mathbf{0})$ instead of just working with A ?

Answer: The augmented matrix tracks the **right-hand side** of the equation!

For $A\mathbf{x} = \mathbf{b}$, we need to track both:

- The coefficients (left side): $\underbrace{A}_{m \times n}$
- The target values (right side): $\underbrace{\mathbf{b}}_{m \times 1}$

Augmented matrix: $\underbrace{(A \mid \mathbf{b})}_{m \times (n+1)}$ keeps them together!

For null space: $\mathbf{b} = \mathbf{0}$, so we use $\underbrace{(A \mid \mathbf{0})}_{m \times (n+1)}$.

Connection: Null Space vs Column Space

Recall from motivation:

Subspace	Natural form	What it records
$\text{Col}(A)$	Constructive $\text{span}\{\text{columns}\}$	All combinations of columns (what we can make)
$\text{Null}(A)$	Descriptive $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$	Relationships between columns (which are redundant)

Connection: If $\text{Null}(A) = \{\mathbf{0}\}$, then:

- No column can be made from others
- All columns are linearly independent
- $\text{Col}(A)$ has dimension = number of columns

Why Cross-Filling Works

Look Back at What We Did

In the example, we decomposed:

$$\underbrace{(A \mid \mathbf{0})}_{4 \times 4} = \underbrace{(R_1 \mid \mathbf{b}_1)}_{\text{Red}} + \underbrace{(R_2 \mid \mathbf{b}_2)}_{\text{Blue}} + \underbrace{(R_3 \mid \mathbf{b}_3)}_{\text{Green}}$$

Each piece gave us a simplified equation:

- $(R_1 \mid \mathbf{b}_1)$ had only x_1, x_3 active (from row 4)
- $(R_2 \mid \mathbf{b}_2)$ had only x_2, x_3 active (from row 3)
- $(R_3 \mid \mathbf{b}_3)$ had only x_3 active (from rows 1,2)

Question: Why does each piece give a simplified equation? What's the underlying principle?

Principle 1: Each Piece = One Equation

Each rank-one piece $\underbrace{(R_i \mid \mathbf{b}_i)}_{m \times (n+1)}$ represents **one equation**.

Why? Rank-one means:

$$(R_i \mid \mathbf{b}_i) = \underbrace{\mathbf{u}}_{m \times 1} \cdot \underbrace{\mathbf{v}^T}_{1 \times (n+1)}$$

where $\mathbf{u}$ = column (normalized pivot column) and $\mathbf{v}^T$ = row (pivot row).

In our red piece:

$$\underbrace{\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{\mathbf{u}} \cdot \underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \end{pmatrix}}_{\text{actually just row 4: } \begin{pmatrix} 1 & 0 & 1 & | & 0 \end{pmatrix}}$$

Principle 1: Each Piece = One Equation (Corrected)

Each rank-one piece gives ONE equation from the **pivot row**.

From our example (Red piece, pivot row 4):

$$\underbrace{(R_1 \mid \mathbf{b}_1)}_{\text{rank-one}} = \text{only row 4 is non-zero: } \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \end{array} \right)$$

This says: $1 \cdot x_1 + 0 \cdot x_2 + 1 \cdot x_3 = 0$, i.e., $\boxed{x_1 + x_3 = 0}$.

From our example (Blue piece, pivot row 3):

$$\underbrace{(R_2 \mid \mathbf{b}_2)}_{\text{rank-one}} = \text{only row 3 is non-zero: } \left(\begin{array}{ccc|c} 0 & 2 & 4 & 0 \end{array} \right)$$

This says: $0 \cdot x_1 + 2 \cdot x_2 + 4 \cdot x_3 = 0$, i.e., $\boxed{2x_2 + 4x_3 = 0}$.

Principle 2: Variables Decrease

As cross-filling proceeds, each piece has **fewer active variables**.

In our example:

Piece	Pivot row equation	Active variables
$(R_1 \mid \mathbf{b}_1)$	$x_1 + x_3 = 0$	x_1, x_3 (2 vars)
$(R_2 \mid \mathbf{b}_2)$	$2x_2 + 4x_3 = 0$	x_2, x_3 (2 vars)
$(R_3 \mid \mathbf{b}_3)$	$x_3 = 0$	x_3 (1 var)

General principle: After eliminating column j_i with pivot, **at least column j_i is gone** from the remainder (possibly more columns eliminated together).

$\Rightarrow$ Each successive piece has **at least one fewer variable**.

Principle 3: V Matrix Collects Simplified Equations

Stack all the pivot rows to form the **V** matrix:

$$V = \begin{pmatrix} \text{— row from } (R_1 \mid \mathbf{b}_1) \text{—} \\ \text{— row from } (R_2 \mid \mathbf{b}_2) \text{—} \\ \vdots \\ \text{— row from } (R_r \mid \mathbf{b}_r) \text{—} \end{pmatrix}$$

In our example (3 pivot rows):

$$V = \begin{pmatrix} 1 & 0 & 1 & | & 0 \\ 0 & 2 & 4 & | & 0 \\ 0 & 0 & 2 & | & 0 \end{pmatrix}$$

Each row = one simplified equation!

Principle 4: Left Cancellation (Why V is Equivalent to A)

Key question: Why is solving $V\mathbf{x} = \mathbf{0}$ the same as solving $A\mathbf{x} = \mathbf{0}$?

Answer: Because $(A \mid \mathbf{0}) = (R_1 \mid \mathbf{b}_1) + \cdots + (R_r \mid \mathbf{b}_r)$.

This means:

$$\underbrace{A}_{m \times n} \mathbf{x} = \underbrace{R_1 + \cdots + R_r}_{m \times n} \mathbf{x}$$

Each R_i is rank-one: $R_i = \mathbf{u}_i \mathbf{v}_i^T$ where $\mathbf{v}_i^T$ is the i -th row of V .

$$R_i \mathbf{x} = \mathbf{u}_i (\mathbf{v}_i^T \mathbf{x})$$

$$\text{So: } A\mathbf{x} = \sum_{i=1}^r \mathbf{u}_i (\mathbf{v}_i^T \mathbf{x}) = \underbrace{U}_{m \times r} \underbrace{V}_{r \times n} \mathbf{x}$$

Left Cancellation: $UV\mathbf{x} = \mathbf{0} \iff V\mathbf{x} = \mathbf{0}$

We have: $\underbrace{A}_{m \times n} = \underbrace{U}_{m \times r} \underbrace{V}_{r \times n}$

Question: Does $A\mathbf{x} = \mathbf{0}$ imply $V\mathbf{x} = \mathbf{0}$?

Answer: YES! By left cancellation:

$$UV\mathbf{x} = \mathbf{0}$$

$$\iff UV\mathbf{x} = U\mathbf{0} \quad (\text{since } U\mathbf{0} = \mathbf{0})$$

$$\iff V\mathbf{x} = \mathbf{0} \quad (\text{left cancel } U \text{ if columns of } U \text{ are linearly independent})$$

Why can we cancel U ? Because U has rank r (full column rank) $\Rightarrow$ its columns are linearly independent!

Explicit U and V in Our Example

From our cross-filling:

$$U = \begin{pmatrix} | & | & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$V = \begin{pmatrix} - & \mathbf{v}_1^T & - \\ - & \mathbf{v}_2^T & - \\ - & \mathbf{v}_3^T & - \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

Check: $UV = A$ and $\underbrace{V}_{3 \times 3} \mathbf{x} = \mathbf{0}$ gives the simplified system!

Summary: Why Cross-Filling Works

Four key principles:

1. **Each rank-one piece = one equation** (from the pivot row)
2. **Variables decrease** (at least one variable eliminated per step)
3. **V matrix collects all simplified equations** (stack pivot rows)
4. **Left cancellation:** $A = UV$ and $UV\mathbf{x} = \mathbf{0} \iff V\mathbf{x} = \mathbf{0}$

Result: Solving $\underbrace{A}_{m \times n} \mathbf{x} = \mathbf{0}$ reduces to solving $\underbrace{V}_{r \times n} \mathbf{x} = \mathbf{0}$ where V has **simpler structure** (upper triangular with decreasing variables).

$\Rightarrow$ Backward substitution becomes easy!

General Cross-Filling Algorithm (Formalized)

Input: $m \times n$ matrix $\underbrace{A}_{m \times n}$ and target $\underbrace{\mathbf{b}}_{m \times 1}$

Output: Solution to $A\mathbf{x} = \mathbf{b}$ (if exists)

Steps:

1. Form augmented matrix $\underbrace{(A \mid \mathbf{b})}_{m \times (n+1)}$
2. Cross-fill: $(A \mid \mathbf{b}) = (R_1 \mid \mathbf{b}_1) + \cdots + (R_r \mid \mathbf{b}_r)$
3. Collect pivot rows into V matrix (size $r \times (n+1)$)
4. Check each non-pivot row:
 - If row = $\begin{pmatrix} 0 & \cdots & 0 & \mid & 0 \end{pmatrix} \Rightarrow \text{OK}$
 - If row = $\begin{pmatrix} 0 & \cdots & 0 & \mid & c \end{pmatrix}$ with $c \neq 0 \Rightarrow \text{NO solution!}$
5. If all non-pivot rows are OK, solve $V\mathbf{x} = \mathbf{b}_V$ backwards

Solving General Equations $Ax = b$

New Example: Non-Zero Right-Hand Side

Setup: Solve $\underbrace{A}_{3 \times 4} \mathbf{x} = \underbrace{\mathbf{b}}_{3 \times 1}$ where

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 1 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 11 \\ 16 \end{pmatrix}$$

Difference from before: Now $\mathbf{b} \neq \mathbf{0}$, so we're looking for **actual solutions** (not just relationships).

Method: Same cross-filling, but track the right-hand side!

Cross-Filling with $b \neq 0$

Form augmented matrix and cross-fill (details similar to before):

$$(A \mid \mathbf{b}) = \underbrace{(R_1 \mid \mathbf{b}_1)}_{\text{pivot (1,1)}} + \underbrace{(R_2 \mid \mathbf{b}_2)}_{\text{pivot (2,3)}}$$

Collect pivot rows into V :

$$V = \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 5 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right)$$

Equations from V :

$$x_1 + 2x_2 + 0x_3 + 1x_4 = 5$$

$$0x_1 + 0x_2 + 1x_3 + 0x_4 = 1$$

Identify Pivot and Free Variables

From the V matrix:

$$V = \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 5 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right)$$

Pivot variables: x_1 (from row 1), x_3 (from row 2)

Free variables: x_2, x_4 (columns without pivots)

Strategy: Set free variables to arbitrary values, solve for pivot variables.

Solve Backwards: Find One Particular Solution

Set free variables to convenient values: $x_2 = 0, x_4 = 0$.

From equation 2: $x_3 = 1$

From equation 1: $x_1 + 2(0) + 0(1) + 1(0) = 5 \Rightarrow x_1 = 5$

One particular solution:

$$\mathbf{x}_p = \begin{pmatrix} 5 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\text{Verify: } A\mathbf{x}_p = \begin{pmatrix} 5 \\ 11 \\ 16 \end{pmatrix} = \mathbf{b} \quad \checkmark$$

General Solution: Add Null Space

Question: Are there other solutions besides $\mathbf{x}_p$?

Answer: YES! Any $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$ where $\mathbf{x}_n \in \text{Null}(A)$.

Why?

$$\begin{aligned} A(\mathbf{x}_p + \mathbf{x}_n) &= A\mathbf{x}_p + A\mathbf{x}_n \\ &= \mathbf{b} + \mathbf{0} = \mathbf{b} \quad \checkmark \end{aligned}$$

To find all solutions:

1. Find one particular solution $\mathbf{x}_p$ (set free vars = 0)
2. Find basis for $\text{Null}(A)$ (set each free var = 1, others = 0)
3. General solution: $\mathbf{x} = \mathbf{x}_p + c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots$

Finding Null Space Basis (Concrete Example)

From V :

$$V = \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{array} \right)$$

Free variables: x_2, x_4 . We need 2 basis vectors.

Vector 1: Set $x_2 = 1, x_4 = 0$

$$x_3 = 0 \quad (\text{from row 2})$$

$$x_1 + 2(1) + 0(0) + 1(0) = 0 \quad \Rightarrow \quad x_1 = -2 \quad (\text{from row 1})$$

$$\mathbf{v}_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

Finding Null Space Basis (Continued)

Vector 2: Set $x_2 = 0, x_4 = 1$

$$x_3 = 0 \quad (\text{from row 2})$$

$$x_1 + 2(0) + 0(0) + 1(1) = 0 \quad \Rightarrow \quad x_1 = -1 \quad (\text{from row 1})$$

$$\mathbf{v}_2 = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Basis for Null(A):

$$\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Complete Solution Set

General solution to $Ax = b$:

$$\mathbf{x} = \underbrace{\begin{pmatrix} 5 \\ 0 \\ 1 \\ 0 \end{pmatrix}}_{\mathbf{x}_p} + c_1 \underbrace{\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}}_{\mathbf{v}_1} + c_2 \underbrace{\begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{\mathbf{v}_2}$$

where $c_1, c_2 \in \mathbb{R}$ are arbitrary.

Geometric picture: The solution set is an **affine subspace**:

$$\text{Solutions} = \mathbf{x}_p + \text{Null}(A)$$

(A 2-dimensional plane in $\mathbb{R}^4$ passing through $\mathbf{x}_p$)

Existence: When Does $Ax = b$ Have Solutions?

Look at the cross-filling decomposition:

$$(A \mid \mathbf{b}) = (R_1 \mid \mathbf{b}_1) + \cdots + (R_r \mid \mathbf{b}_r)$$

Two cases for non-pivot rows:

- **Case 1:** Non-pivot row = $\begin{pmatrix} 0 & \cdots & 0 & \mid & 0 \end{pmatrix}$
This is OK! Just says $0 = 0$ (always true).
- **Case 2:** Non-pivot row = $\begin{pmatrix} 0 & \cdots & 0 & \mid & c \end{pmatrix}$ with $c \neq 0$
This is CONTRADICTION! Says $0 = c$ where $c \neq 0$ (impossible).

Conclusion: Solution exists $\iff$ no Case 2 rows appear.

Visual: Existence Cases

Case 1: Solution Exists

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Non-pivot row: $0 = 0$ ✓
Pivot rows colored

Case 2: No Solution

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{array} \right)$$

Non-pivot row: $0 = 3$ ✗
Contradiction highlighted

Key: If $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A) + 1$, then Case 2 occurs!

Uniqueness: When Is the Solution Unique?

Suppose $A\mathbf{x} = \mathbf{b}$ has at least one solution $\mathbf{x}_p$.

Question: Is this the **only** solution?

Answer: Depends on $\text{Null}(A)$!

- If $\text{Null}(A) = \{\mathbf{0}\}$, then $\mathbf{x} = \mathbf{x}_p + \mathbf{0} = \mathbf{x}_p$ is unique.
- If $\text{Null}(A) \neq \{\mathbf{0}\}$, then $\mathbf{x} = \mathbf{x}_p + c_1\mathbf{v}_1 + \cdots$ gives infinitely many solutions.

Conclusion: Solution is unique $\iff \text{Null}(A) = \{\mathbf{0}\}$.

Applications: Existence, Uniqueness, Null Space

Application 1: Finding Null Space Basis (Summary)

Proposition

To find a basis for $\text{Null}(\underbrace{A}_{m \times n})$:

1. Cross-fill $(A \mid \mathbf{0})$ to get V matrix
2. Identify free variables (columns without pivots)
3. For each free variable: set it = 1, others = 0, solve for pivot variables
4. Resulting vectors form a basis for $\text{Null}(A)$

Dimension: $\dim(\text{Null}(A)) = \text{number of free variables.}$

Application 2: Existence Criterion

Proposition (Existence)

$$\underbrace{A}_{m \times n} \mathbf{x} = \underbrace{\mathbf{b}}_{m \times 1} \text{ has a solution } \iff \underbrace{\text{rank}(A \mid \mathbf{b})}_{m \times (n+1)} = \underbrace{\text{rank } A}_{m \times n}.$$

Interpretation:

- $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A)$ means $\mathbf{b}$ doesn't add new information
 $\Rightarrow \mathbf{b}$ can be made from columns of A
- $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A) + 1$ means $\mathbf{b}$ adds new dimension
 $\Rightarrow \mathbf{b}$ is NOT in $\text{Col}(A)$, contradiction appears

Application 3: Uniqueness Criterion

Proposition (Uniqueness)

If $\underbrace{A}_{m \times n} \mathbf{x} = \underbrace{\mathbf{b}}_{m \times 1}$ has solutions, then the solution is **unique** $\iff$
 $\text{Null}(A) = \{\mathbf{0}\}$.

Equivalent conditions for uniqueness:

- $\text{Null}(A) = \{\mathbf{0}\}$
- $\text{rank}(A) = n$ (all columns are pivot columns)
- No free variables
- Columns of A are linearly independent

Application 4: Structure of Solution Set

Theorem (General Solution Structure)

If $\underbrace{A}_{m \times n} \mathbf{x} = \underbrace{\mathbf{b}}_{m \times 1}$ has solutions, **all solutions** have the form:

$$\mathbf{x} = \underbrace{\mathbf{x}_p}_{\text{particular solution}} + \underbrace{\mathbf{x}_n}_{\in \text{Null}(A)}$$

where $\mathbf{x}_p$ is any particular solution and $\mathbf{x}_n$ is arbitrary.

Geometric picture: Solution set = $\mathbf{x}_p + \text{Null}(A)$ (affine subspace).

Proof idea: If $\mathbf{x}_1, \mathbf{x}_2$ are both solutions, then $\mathbf{x}_1 - \mathbf{x}_2 \in \text{Null}(A)$.

Dimension Formula: Rank-Nullity Theorem

Counting Pivot and Free Variables

From cross-filling an $m \times n$ matrix $\underbrace{A}_{m \times n}$:

- Number of pivot columns = $\text{rank}(A)$
- Number of free variables = $\dim(\text{Null}(A))$
- Total columns = n

Every column is either pivot or non-pivot:

$$\text{rank}(A) + \dim(\text{Null}(A)) = n$$

This is the **Rank-Nullity Theorem**.

Visual Example: Rank-Nullity

From our example (A is 3×4):

$$\begin{array}{cccc} & \text{pivot} & \text{free} & \text{pivot} & \text{free} \\ \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 0 & 1 \\ \hline 2 & 4 & 1 & 2 \\ \hline 3 & 6 & 1 & 3 \\ \hline \end{array} \right) \end{array}$$

Pivot columns (rank = 2)

Non-pivot columns (nullity = 2)

$$\underbrace{2}_{\text{rank}} + \underbrace{2}_{\text{nullity}} = \underbrace{4}_n$$

Rank-Nullity Theorem

Theorem (Rank-Nullity Theorem)

For any $m \times n$ matrix $\underbrace{A}_{m \times n}$:

$$\text{rank}(A) + \dim(\text{Null}(A)) = n$$

Proof idea: Every column is either:

- A pivot column (contributes to rank), or
- A non-pivot column (contributes one free variable to nullity)

Total = n columns.

Preview: Null Space and Row Space Orthogonality

Natural Orthogonality

Theorem

$$\text{Null}(\underbrace{A}_{m \times n}) \perp \text{Row}(\underbrace{A}_{m \times n})$$

Every vector in $\text{Null}(A)$ is orthogonal to every vector in $\text{Row}(A)$.

Proof: Let $\mathbf{x} \in \text{Null}(A)$ and $\mathbf{r}_i = i$ -th row of A .

By definition: $A\mathbf{x} = \mathbf{0}$ means $\mathbf{r}_i \cdot \mathbf{x} = 0$ for all i .

$\Rightarrow \mathbf{x} \perp \text{every row} \Rightarrow \mathbf{x} \perp \text{Row}(A)$.



Coffee Shop Interpretation

If $\underbrace{A}_{m \times n} = \text{"drinks} \rightarrow \text{ingredients"}:$

- **Row space:** All ingredient combinations from drinks
- **Null space:** Relationships between drinks (which can be made from others)

Perpendicular: A relationship (which drinks are equivalent) has nothing in common with actual ingredient requirements!

Next lecture: Four fundamental subspaces and complete orthogonal structure.

Summary

What We Achieved Today

1. Concrete Example First:

- Started with coffee shop null space problem
- Solved completely using cross-filling before defining theory

2. Extracted Theory:

- Null space definition from the example
- Cross-filling method: each $(R_i \mid \mathbf{b}_i) =$ one equation
- Variables decrease $\rightarrow$ backward solving
- V matrix and left cancellation

3. Applied to General Equations:

- Existence: $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A) \iff$ solution exists
- Uniqueness: $\text{Null}(A) = \{\mathbf{0}\} \iff$ unique solution
- General solution structure: $\mathbf{x} = \mathbf{x}_p + \text{Null}(A)$

What We Achieved Today (continued)

4. Rank-Nullity:

$$\text{rank}(A) + \dim(\text{Null}(A)) = n$$

Visual: pivot columns + non-pivot columns = all columns

5. Orthogonality Preview: $\text{Null}(A) \perp \text{Row}(A)$

Next lecture: Four fundamental subspaces and their complete orthogonal structure!